

Climate Change Twitter Dataset Analysis Report

1. Introduction

This project presents an end-to-end analysis of the Climate Change Twitter Dataset using PostgreSQL and Power BI. The goal of the analysis was to help a climate research organisation understand how public opinion on climate change has changed over time, and to identify the topics and locations associated with the most active, divisive, or aggressive discussions.

The project followed a full analytics workflow, beginning with raw data ingestion into PostgreSQL, followed by data quality checks, cleaning, transformation, descriptive and diagnostic analysis, and finally dashboard development in Power BI.

2. Data Understanding

The dataset contains over 15 million climate-related tweets. Each record represents a tweet and includes information about when it was posted, its geographic coordinates, topic category, sentiment score, stance on climate change, author gender, temperature deviation, and aggressiveness label.

In addition to the Twitter dataset, a disaster dataset was incorporated as a supporting dataset. While both datasets differ in structure and granularity, they share a common time dimension (year), which allows for comparative analysis of real-world climate events and online discourse trends.

The main columns in the main dataset are:

Column	Description
created_at	Timestamp showing when the tweet was posted
id	Unique tweet identifier
lng	Longitude of the tweet location
lat	Latitude of the tweet location
topic	Climate-related discussion category
sentiment	Sentiment score from -1 to 1
stance	Tweet stance: believer, denier, or neutral
gender	Author gender: male, female, or undefined
temperature_avg	Temperature deviation at the time/location of the tweet
aggressiveness	Whether the tweet is aggressive or not

Although the task description stated that the dataset spans from 2008 to 2022, SQL exploration showed that the actual data available ranges from 2006 to 2019. Therefore, the analysis was based on the available data in the dataset.

3. Data Ingestion

The dataset was imported into PostgreSQL as a raw table named `climate_tweets_raw`. All columns were initially stored as text to avoid import errors caused by missing values, inconsistent formats, or type conversion issues during ingestion.

After import, the row count was checked and the first 50 records were inspected to confirm that the data loaded correctly.

4. Data Quality Checks

Several data quality checks were performed before transformation.

First, missing coordinate values were checked. The analysis showed that about 10.4 million records had missing latitude and longitude values, while about 5.3 million records had usable coordinates. Since both latitude and longitude are needed for location-based analysis, records without coordinates were retained for general analysis but excluded from geographic analysis.

Gender values were also checked. The dataset was male-dominated, and more than 500,000 records had undefined gender values. These undefined values were later standardized to “Unknown” in the cleaned view.

Duplicate checks were performed using the tweet id column. No duplicate tweet IDs were found, indicating that each tweet appeared once in the dataset.

Sentiment scores were checked to ensure they fell within the expected range of -1 to 1. No out-of-range sentiment values were found. Missing sentiment checks also returned no missing sentiment values.

Initial exploration also showed that believers made up the largest stance category, while most tweets were not aggressive.

5. Data Cleaning and Preparation

A cleaned view named `vw_cleaned_tweets` was created in PostgreSQL. This view converted key columns into appropriate data types and standardized inconsistent values.

The following transformations were applied:

- `created_at` was converted to timestamp
- `lng` and `lat` were converted to numeric coordinates
- `sentiment` was converted to numeric format
- `temperature_avg` was converted to numeric format
- `gender = undefined` was changed to Unknown
- `tweet_year` was extracted from `created_at`

This cleaned view served as the foundation for all subsequent analysis views and Power BI visuals.

6. Data Wrangling and Transformation

Because the main Twitter dataset was provided as a single table, no direct joins were required for the main analysis. Instead, PostgreSQL was used to transform the raw dataset into meaningful analysis-ready views.

The views created include:

View	Purpose
vw_cleaned_tweets	Cleaned and typed version of the raw dataset
vw_sentiment_trend	Yearly tweet count and average sentiment
vw_stance_trend	Stance distribution by year
vw_topic_aggression	Topic-level aggression analysis
vw_topic_stance	Topic-level stance analysis
vw_location_sentiment	Location-based tweet volume and sentiment
vw_topic_stance_year	Topic, stance, and year analysis for interactive filtering

Indexes were also created on important columns such as created_at, stance, topic, and aggressiveness to improve query performance on the large dataset.

7. Descriptive Analytics

Descriptive analytics were used to summarize the dataset and understand its key characteristics.

The sentiment trend view showed yearly average sentiment and tweet volume. This revealed that tweet activity increased significantly over time, with 2018 recording the highest number of tweets. The average sentiment was slightly negative overall, suggesting that climate discussions were generally mixed but leaned slightly negative.

Stance distribution showed that believers dominated the dataset. Neutral tweets formed a noticeable portion, while deniers were the smallest group.

Aggressiveness distribution showed that most tweets were not aggressive, but aggressive tweets still represented a significant share of the discussion. In the dashboard, aggressive tweets accounted for about 28.7% of total tweets.

8. Diagnostic Analytics

Diagnostic analytics were used to investigate patterns behind the trends.

The topic and aggression analysis showed that major topics such as global stance, weather extremes, and human intervention dominated climate discussions. Although most tweets were not aggressive, a substantial number of aggressive tweets existed across major topics.

The topic and stance analysis showed that belief in climate change dominated across most topics. However, the presence of believer, denier, and neutral stances across topic categories showed that climate discourse is still contested and not completely one-sided.

The aggression percentage by topic helped identify topics with higher proportions of aggressive tweets, making it easier to understand which topics are more likely to trigger confrontational discussions.

Geographic analysis was performed using only records with valid coordinates. Coordinates were rounded to one decimal place to group nearby locations and improve map performance. This helped identify locations with high tweet activity and compare their average sentiment.

To better understand the drivers of climate discourse, disaster occurrence trends were analysed alongside Twitter activity using a shared time dimension. Although the two datasets were not directly joined due to the absence of a common key, they were aligned by year to enable comparative analysis.

The comparison revealed that while tweet volume increased significantly over time, disaster occurrences remained relatively stable. This suggests that the growth in climate-related discussions is not solely driven by the frequency of disasters, but also by increasing public awareness, media coverage, and global engagement with climate issues.

Therefore, the disaster dataset was used as contextual support rather than a directly integrated data source, helping to explain trends observed in public sentiment and engagement.

9. Dashboard Explanation

The Power BI dashboard was structured into four pages.

Page 1: Executive Summary

This page provides a high-level overview of the dataset. It includes KPI cards for total tweets, average sentiment, aggressive tweets, and aggression percentage.

It also includes:

- Sentiment trend over time
- Tweet volume over time
- Disaster occurrence trend

This page helps leadership quickly understand the scale, tone, and growth of climate-related conversations.

Page 2: Climate Stance Analysis

This page focuses on how public stance toward climate change changes over time.

It includes:

- 100% stacked chart showing stance distribution over time
- Donut chart showing overall stance distribution
- Tweet volume by stance over time

The analysis shows that believers dominate climate discourse, while neutral and denier groups remain smaller.

Page 3: Topic Analysis and Aggression Patterns

This page identifies topics driving discussion and aggression.

It includes:

- Climate stance distribution across top topics
- Aggressive vs non-aggressive tweet distribution
- Aggression rate by topic

This page helps identify which topics are more likely to generate aggressive or divisive discourse.

Page 4: Geographic Analysis

This page shows the geographic spread of climate tweets.

It includes:

- A map showing tweet concentration by location
- A table showing top locations by tweet volume and sentiment

This page helps identify where climate-related conversations are concentrated.

10. Key Findings

The analysis produced the following major findings:

- The dataset contains about 15.79 million tweets.
- The actual dataset range is 2006 to 2019, despite the task description stating 2008 to 2022.
- Tweet volume increased significantly over time, with 2018 recording the highest tweet activity.
- Average sentiment was slightly negative, showing that climate discourse is generally mixed but leans negative.
- Believers form the largest stance group, showing that most users support the belief in climate change.
- Deniers form the smallest stance group, but denial is still present across the dataset.
- About 28.7% of tweets were aggressive, indicating a notable level of confrontational discourse.
- Major topics such as global stance, weather extremes, and human intervention dominate the conversation.
- Some topics show higher aggression rates, suggesting that certain climate discussions are more polarizing.
- A large number of records lacked coordinates, limiting the completeness of geographic analysis.

11. Recommendations

Based on the analysis, the following recommendations are proposed:

Climate organisations should focus communication efforts on high-volume topics such as global stance, weather extremes, and human intervention because these topics attract the most public attention.

Topics with higher aggression rates should be monitored more closely. These areas may require clearer messaging, moderation, or educational campaigns to reduce misinformation and hostile debate.

Since a large portion of users are neutral, communication strategies should target this group with accessible and evidence-based content. Neutral users may be more open to persuasion than deniers.

Geographic gaps in tweet activity suggest that public discourse is uneven across regions. Climate organisations should improve outreach in underrepresented areas to ensure broader participation in climate conversations.

Social media campaigns should consider both sentiment and aggressiveness, not just tweet volume. A high number of tweets does not always indicate positive engagement.

12. Limitations

The analysis has some limitations.

First, more than 10 million records had missing coordinate values, limiting the completeness of geographic analysis.

Second, sentiment scores may not fully capture sarcasm, irony, or complex emotional tone in tweets.

Third, Twitter data may not represent the entire global population, since social media participation differs by region, internet access, and demographics.

Finally, the actual date range of the dataset differed from the task description, so the analysis was based on the years available in the imported data.

13. Conclusion

This project successfully demonstrates an end-to-end analytics workflow using PostgreSQL and Power BI. PostgreSQL was used to store, clean, transform, and structure the data into reusable analytical views, while Power BI was used to create an interactive dashboard for exploring trends, stance, topics, aggression, and geography.

The findings show that climate discourse has grown significantly over time and is largely dominated by believers. However, a substantial amount of aggressive discourse exists, especially around certain topics. This highlights the need for clearer communication, better public engagement, and focused monitoring of polarizing climate discussions.

14. References

- Climate Change Twitter Dataset provided for the Stage 4 task
- Disaster dataset provided as supporting context
- PostgreSQL Documentation
- Microsoft Power BI Documentation